Fixed Smooth Targets Are Poorly Approximated by Uniformly Learnable Function Classes

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Abstract

We answer the fixed-smoothness question in arXiv:2005.10807 in the natural high-dimensional regime. If the unit ball of a Banach function space X admits a uniform Monte-Carlo integration rate $n^{-1/2}$, then, for $d > 2k$, the C^k unit ball has L^2 approximation width relative to tB_X bounded below by $t^{-2k/(d-2k)}$. A Baire-category argument produces one fixed C^k target with the corresponding limsup obstruction for squared error. The proof uses disjoint smooth bumps which avoid the small averaging balls needed to make the integration operators continuous on L^2.

1 Statement

Work on the flat torus $Q = \mathbb{T}^d$, identified with $[0,1]^d$ with periodic boundary conditions, and let P be normalized Lebesgue measure. For an integer $k \geq 1$, equip $C^k(Q)$ with

$$\|g\|_{C^k} = \max_{|\alpha|\leq k} \|D^\alpha g\|_\infty.$$

Let X be a Banach space continuously embedded in $L^2(Q)$. Assume its unit ball satisfies the uniform Monte-Carlo estimate

$$\mathbb{E} \sup_{\|f\|_X \leq 1} \left| \frac{1}{n} \sum_{i=1}^n f(U_i) - Pf \right| \leq C_X n^{-1/2}, \tag{1}$$

where the U_i are independent and uniform on Q. This is the hypothesis used in arXiv:2005.10807; in particular it holds for its Barron and tree-like neural-network spaces.

Theorem 1. *Assume (1) and $d > 2k$. There are constants $c, t_0 > 0$ such that*

$$\sup_{\|g\|_{C^k}\leq 1} \inf_{\|f\|_X \leq t} \|g - f\|_{L^2(Q)} \geq c\, t^{-\frac{2k}{d-2k}} \qquad (t \geq t_0). \tag{2}$$

There also exists a single $g \in C^k(Q)$, with $\|g\|_{C^k} \leq 1$, such that for every $\eta > 4k/(d-2k)$,

$$\limsup_{t\to\infty} t^\eta \inf_{\|f\|_X \leq t} \|g - f\|^2_{L^2(Q)} = \infty. \tag{3}$$

For each fixed k, the hypothesis $d > 2k$ holds in all sufficiently large dimensions. Thus Theorem 1 proves the source paper's conjectured C^k replacement in precisely its intended fixed-smoothness, high-dimensional regime.

2 Smooth empirical discrepancy

The source asks about the integral probability metric generated by functions which are 1-Lipschitz and have bounded k-th derivative. Modulo constants, this class is equivalent, up to constants depending on d, k and the fixed derivative bound, to a bounded subset of $C^k(Q)$. One direction is immediate; the other follows from the standard finite-difference interpolation inequalities controlling intermediate derivatives by the first and k-th derivatives. Constants do not affect integration against a signed measure of total mass zero.

Kloeckner proved that for every probability measure μ on a bounded d-dimensional domain,

$$\mathbb{E}\|\mu_n - \mu\|_{(C^k)^*} \lesssim_{d,k} \begin{cases} n^{-k/d}, & d > 2k, \\ (\log n) n^{-1/2}, & d = 2k, \\ n^{-1/2}, & d < 2k. \end{cases} \tag{4}$$

For uniform measure in the high-dimensional regime the first rate is sharp. We record an elementary lower bound in a form that will also survive mollification.

For $\varepsilon > 0$ and a point set $S = (x_1, \ldots, x_n) \subset Q$, define

$$A_{S,\varepsilon} f = \frac{1}{n} \sum_{i=1}^{n} \frac{1}{|B_\varepsilon|} \int_{B_\varepsilon(x_i)} f(y)\, dy, \qquad Af = Pf. \tag{5}$$

The balls are periodic Euclidean balls.

Lemma 2 (mollified smooth discrepancy)**.** *For every $d, k \geq 1$ there are $a, c > 0$ such that, with $\varepsilon_n = an^{-1/d}$, every n-point set $S \subset Q$ satisfies*

$$\|A_{S,\varepsilon_n} - A\|_{(C^k)^*} \geq cn^{-k/d}. \tag{6}$$

The constants depend only on d and k.

Proof. Fix a nonnegative $\psi \in C_c^\infty((-1/4, 1/4)^d)$ with $b := \int \psi > 0$. Let $m = \lceil 10n^{1/d} \rceil$, $h = m^{-1}$, and use the periodic grid of m^d centers z_j. Take $a = 1/100$.

Call z_j bad if

$$\|z_j - x_i\|_\infty \leq \varepsilon_n + h/4$$

for some sample point x_i. For a fixed x_i, the number of grid centers in this cube is at most

$$(2m\varepsilon_n + 3)^d \leq 4^d,$$

because $m\varepsilon_n \leq 11/100$. Hence there are at most $4^d n$ bad centers, while $m^d \geq 10^d n$. A fixed positive fraction of all centers is therefore good.

Let $M_k = \max_{|\alpha| \leq k} \|D^\alpha \psi\|_\infty$ and set

$$g_S(x) = M_k^{-1} h^k \sum_{z_j \text{ good}} \psi\left(\frac{x - z_j}{h}\right).$$

The supports are disjoint. For $|\alpha| \leq k$,

$$\|D^\alpha g_S\|_\infty \leq M_k^{-1} h^{k-|\alpha|} \|D^\alpha \psi\|_\infty \leq 1,$$

so $\|g_S\|_{C^k} \leq 1$. By the definition of good centers, the support of g_S misses every ball $B_{\varepsilon_n}(x_i)$, and consequently $A_{S,\varepsilon_n} g_S = 0$. On the other hand,

$$Ag_S = M_k^{-1} b \,\#\{j : z_j \text{ good}\}\, h^{d+k} \geq c_{d,k} h^k \geq c'_{d,k} n^{-k/d}.$$

Changing the sign of g_S if necessary proves (6). $\square$

Taking $\varepsilon = 0$ in the same construction gives, for uniform P and every empirical P_n, the deterministic lower bound $\|P_n - P\|_{(C^k)^*} \gtrsim n^{-k/d}$. Together with (4), this answers the source's smooth empirical-rate question sharply when $d > 2k$.

3 Simultaneous operator estimates

We next choose point sets for which the lower bound of Lemma 2 coexists with the fast estimate on X and a uniform bound on L^2.

Lemma 3. *There are point sets $S_n \subset Q$, $|S_n| = n$, for which, writing $T_n = A_{S_n,\varepsilon_n} - A$,*

$$\|T_n\|_{X^*} \le C_1 n^{-1/2}, \tag{7}$$

$$\|T_n\|_{(C^k)^*} \ge c_1 n^{-k/d}, \tag{8}$$

$$\|T_n\|_{(L^2)^*} \le C_2. \tag{9}$$

Proof. Choose $S = (U_1, \dots, U_n)$ randomly. Averaging first over the ball variable and then using translation invariance of P, (1) gives

$$\mathbb{E}\|A_{S,\varepsilon_n} - A\|_{X^*} \le C_X n^{-1/2}.$$

Notice that this does not require the X-norm to be translation invariant: for every fixed shift, $(U_1 + y, \dots, U_n + y)$ has the original joint law.

Let $K_\varepsilon = |B_\varepsilon|^{-1} 1_{B_\varepsilon}$. The functional T_n has the L^2 density

$$q_S(x) = \frac{1}{n}\sum_{i=1}^{n} K_{\varepsilon_n}(x - U_i) - 1.$$

Independence and centering yield

$$\mathbb{E}\|q_S\|_2^2 = \frac{1}{n}\big(\|K_{\varepsilon_n}\|_2^2 - 1\big) \le \frac{1}{n|B_{\varepsilon_n}|} = C_{d,a},$$

and hence $\mathbb{E}\|T_n\|_{(L^2)^*} \le C_{d,a}^{1/2}$. Markov's inequality shows that with positive probability both desired upper bounds hold with three times their expectations. The lower bound (8) holds for every realization by Lemma 2; selecting one realization proves the claim. $\square$

4 Width separation and a fixed target

We include the short separation argument to make the exponent transparent.

Lemma 4 (functional separation)**.** *Let $X, Y \hookrightarrow Z$ be Banach spaces. Suppose $T_n \in Z^*$ satisfy*

$$\|T_n\|_{X^*} \le C_X n^{-\alpha}, \qquad \|T_n\|_{Y^*} \ge c_Y n^{-\beta}, \qquad \|T_n\|_{Z^*} \le C_Z$$

with $0 < \beta < \alpha$. Then, for all large t,

$$\rho(t) := \sup_{\|y\|_Y \le 1} \operatorname{dist}_Z(y, tB_X) \ge c\, t^{-\beta/(\alpha-\beta)}.$$

Proof. Choose $y_n \in B_Y$ with $|T_n y_n| \geq (c_Y/2)n^{-\beta}$. For $x \in tB_X$,

$$(c_Y/2)n^{-\beta} \leq C_Z \|y_n - x\|_Z + C_X t n^{-\alpha}.$$

Choose an integer $n \asymp t^{1/(\alpha-\beta)}$ large enough that the last term is at most $(c_Y/4)n^{-\beta}$. Taking the infimum over x gives the claim. $\square$

The source's constructive fixed-point lemma imposes the stronger condition $\beta < \alpha/2$. For existence of a fixed hard target, Baire category removes this loss.

Lemma 5 (Baire fixed-target upgrade)**.** *Let $X, Y \hookrightarrow Z$ be Banach spaces, and suppose*

$$\rho(t) = \sup_{\|y\|_Y \leq 1} \operatorname{dist}_Z(y, tB_X) \geq ct^{-q}$$

for all large t. Then for every $\gamma > q$ there exists $y \in B_Y$ such that

$$\limsup_{t\to\infty} t^\gamma \operatorname{dist}_Z(y, tB_X) = \infty.$$

In fact, one y may be chosen to have this property simultaneously for all $\gamma > q$.

Proof. Fix $\gamma > q$ and write $E_t(y) = \operatorname{dist}_Z(y, tB_X)$. Suppose every $y \in Y$ satisfies $\sup_{t\geq 1} t^\gamma E_t(y) < \infty$. Then

$$Y = \bigcup_{M=1}^{\infty} F_M, \qquad F_M = \{y : E_t(y) \leq Mt^{-\gamma} \text{ for all } t \geq 1\}.$$

Each F_M is closed because E_t is continuous on Y. By Baire's theorem, some F_M contains a ball $y_0 + rB_Y$. Put $\delta = r/2$. For every $u \in B_Y$, both y_0 and $y_0 + \delta u$ lie in F_M. Subtracting tB_X-approximants to these two points gives

$$E_{2t}(\delta u) \leq 2Mt^{-\gamma}.$$

The exact scaling identity $E_{2t}(\delta u) = \delta E_{2t/\delta}(u)$ therefore implies

$$\rho(s) \leq C_{M,r,\gamma} s^{-\gamma}$$

for all large s, contradicting $\rho(s) \geq cs^{-q}$.

Thus for each rational $\gamma > q$, the set of points with unbounded $t^\gamma E_t(y)$ is dense G_δ by the same argument applied in every open ball. Intersecting over rational $\gamma > q$ and normalizing a nonzero point gives a single $y \in B_Y$ working for every real $\gamma > q$. $\square$

Proof of Theorem 1. Apply Lemma 4 to the operators of Lemma 3 with

$$Y = C^k(Q), \qquad Z = L^2(Q), \qquad \alpha = \frac{1}{2}, \qquad \beta = \frac{k}{d}.$$

The condition $d > 2k$ is exactly $\beta < \alpha$, and

$$\frac{\beta}{\alpha - \beta} = \frac{k/d}{1/2 - k/d} = \frac{2k}{d - 2k}.$$

This proves (2). Lemma 5 gives one $g \in B_{C^k}$ for which

$$\limsup_{t\to\infty} t^\gamma \inf_{\|f\|_X \leq t} \|g - f\|_2 = \infty \quad \text{for every } \gamma > \frac{2k}{d-2k}.$$

Squaring and putting $\eta = 2\gamma$ proves (3). $\square$

5 Boundary of the result

The exponent transition has a structural meaning. When $d > 2k$, smooth empirical integration is of order $n^{-k/d}$, strictly slower than the $n^{-1/2}$ rate on X, and the separation principle applies. When $d < 2k$, the C^k empirical rate is already $n^{-1/2}$; at $d = 2k$ it is parametric up to a critical logarithm. Thus the present mechanism has no exponent gap at $d \leq 2k$. This is consistent with the source's formulation: k is fixed while dimension becomes large.

The theorem is abstract in X. Substituting the Barron space or the tree-like spaces from arXiv:2005.10807 gives the claimed neural-network consequence immediately. The result does not classify which smooth targets are hard, and it does not address loss functionals such as logistic loss.

6 References

- W. E and S. Wojtowytsch, *Kolmogorov Width Decay and Poor Approximators in Machine Learning: Shallow Neural Networks, Random Feature Models and Neural Tangent Kernels*, Research in the Mathematical Sciences 8, 5 (2021); arXiv:2005.10807.

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